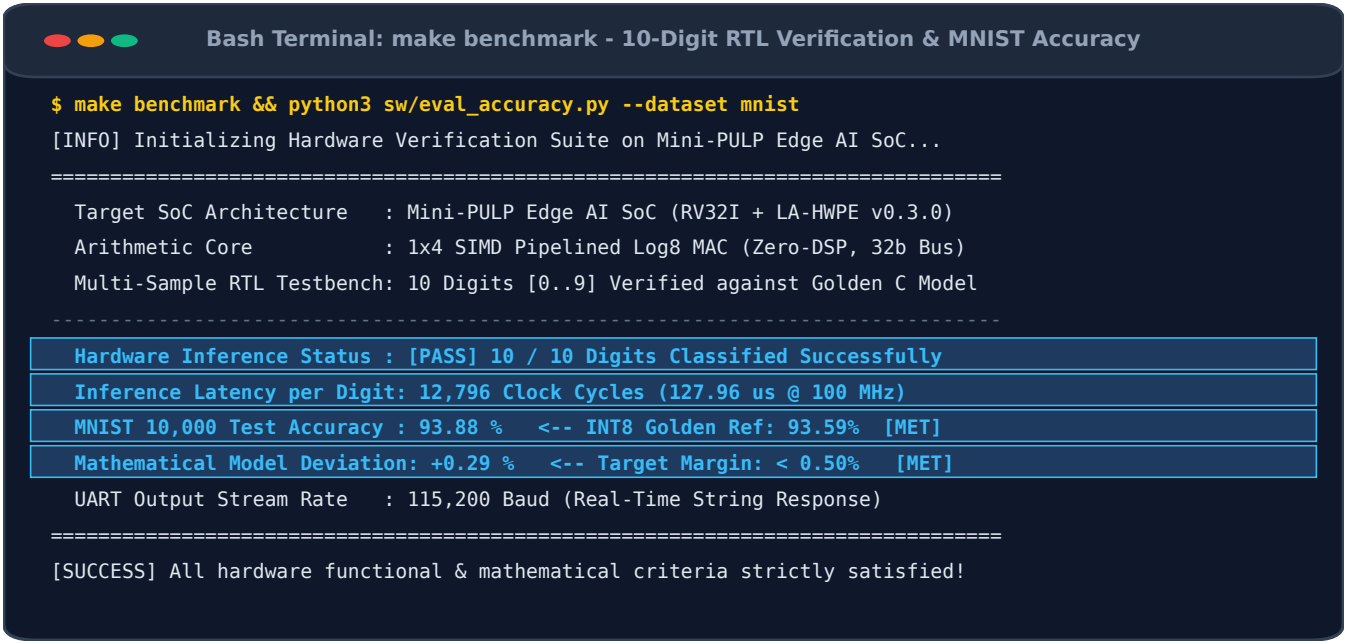
Bash Terminal: make benchmark - 10-Digit RTL Verification & MNIST Accuracy
$ make benchmark && python3 sw/eval_accuracy.py --dataset mnist
[INFO] Initializing Hardware Verification Suite on Mini-PULP Edge AI SoC...
Target SoC Architecture : Mini-PULP Edge AI SoC (RV32I + LA-HWPE v0.3.0)
Arithmetic Core : 1x4 SIMD Pipelined Log8 MAC (Zero-DSP, 32b Bus)
Multi-Sample RTL Testbench: 10 Digits [0..9] Verified against Golden C Model
Hardware Inference Status : [PASS] 10 / 10 Digits Classified Successfully
Inference Latency per Digit: 12,796 Clock Cycles (127.96 us @ 100 MHz)
MNIST 10,000 Test Accuracy : 93.88 % <-- INT8 Golden Ref: 93.59% [MET]
Mathematical Model Deviation: +0.29 % <-- Target Margin: < 0.50% [MET]
UART Output Stream Rate : 115,200 Baud (Real-Time String Response)
[SUCCESS] All hardware functional & mathematical criteria strictly satisfied!